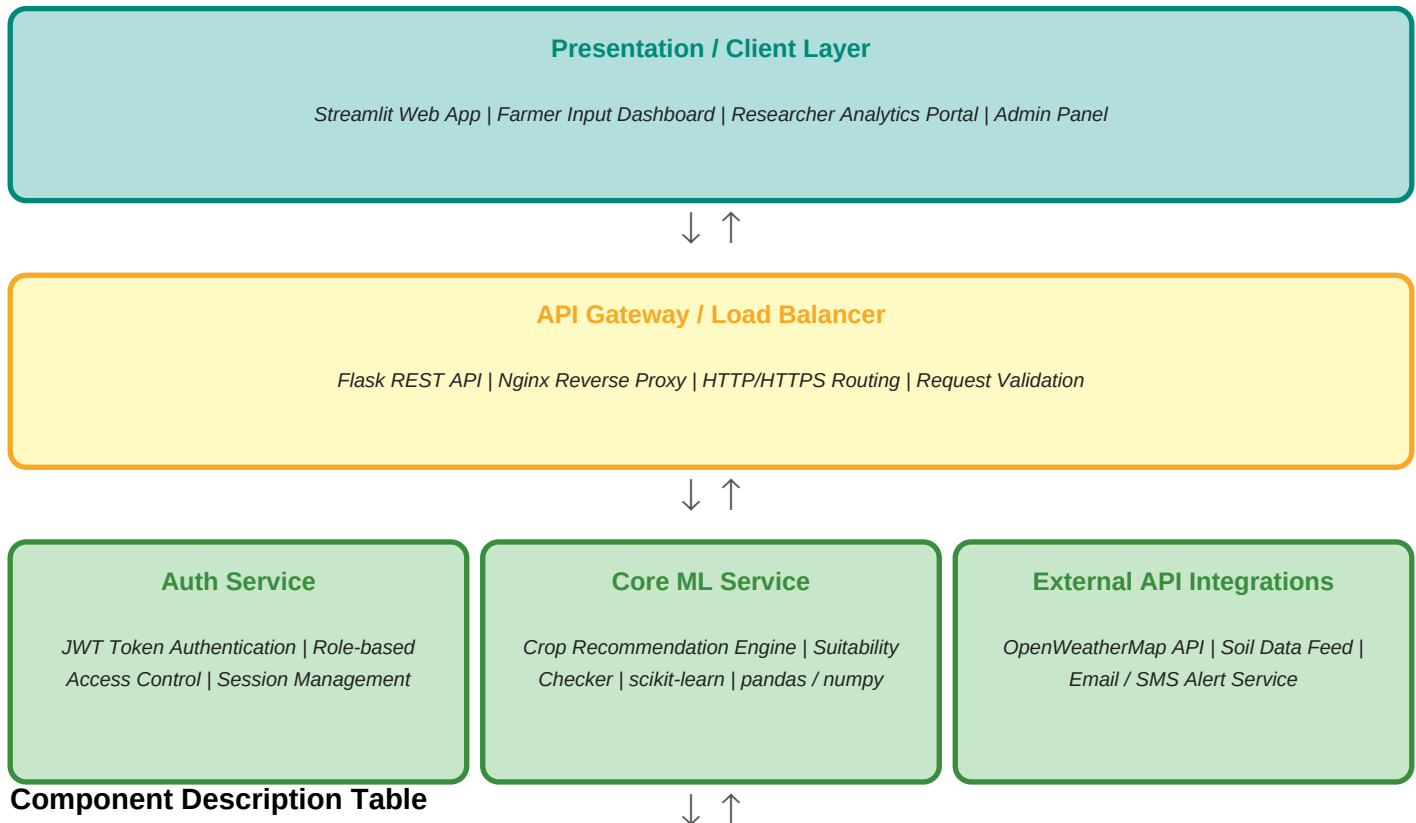


Solution Architecture

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below to map out the Presentation Layer, Application Layer, and Data Layer with specific technologies for OptiCrop.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	The user-facing interface that allows farmers to input soil/environmental parameters and receive crop recommendations. Researchers and policymakers use the analytics portal.	Streamlit, Flask (HTML/CSS), Bootstrap Kaggle / UCI Crop CSV Datasets SQLite / PostgreSQL DB AWS S3 (Model Artifacts)
API Gateway	Acts as the entry point for all client requests. Routes requests to appropriate backend services, handles authentication validation and load distribution.	Flask REST API, Nginx, HTTP/HTTPS

Core ML Service	The heart of OptiCrop — processes soil data (N, P, K, pH, temp, humidity, rainfall) through trained ML models to output crop recommendations and suitability scores.	scikit-learn, pandas, numpy, Random Forest, Decision Tree
Auth Service	Manages user identity and access control. Authenticates farmers, researchers, and admin users with role-based permissions.	JWT Tokens, Flask-Login, OAuth2
External API Integrations	Fetches real-time weather and environmental data to supplement farmer-entered inputs and trigger automated crop advisory alerts.	OpenWeatherMap API, SMTP Email, Twilio SMS
Data / Storage Layer	Stores all crop datasets, trained model artifacts, user records, and historical yield data. Supports both structured and file-based storage.	PostgreSQL, SQLite, AWS S3, CSV (Kaggle/UCI datasets)